MATH0009 Newtonian Mechanics

Year:	2026–2027
Code:	MATH0009
Level:	4 (UG)
Normal student group(s):	UG: Y1 Mathematics programmes
Value:	15 credits (7.5 ECTS credits)
Term:	2
Assessment:	85% examination, 15% coursework. In order to pass the module you must have at least 40% for both the examination mark and the final weighted mark.
Normal Pre-requisites:	MATH0008 (Applied Mathematics) MATH0010 (Mathematical Methods 1)
Lecturer:	Dr R Harris

Course Description and Objectives

This course follows the first term applied mathematics course and gives a comprehensive introduction to Newtonian mechanics. The essential concepts of force, torque, momentum, angular momentum and energy are introduced. This is followed by a thorough coverage of the Newtonian dynamics of point particles, including the classic problem of a central force with the inverse square law. Vector methods, including vector differential equations, are used extensively.

Recommended Texts

1. P Smith and RC Smith, *Mechanics* 2nd ed, (Wiley)
2. CD Collinson, *Introductory Mechanics* (Arnold)
3. M Lunn, *A first course in Mechanics* (Oxford)
4. CD Collinson and T Roper, *Particle Mechanics* (Arnold)

Detailed Syllabus

– Ingredients of mechanics: Force as a vector, moments, couples. Velocity, momentum, angular momentum, acceleration. Newton's Laws. Conservative and non-conservative forces. Energy; energy conservation; conservation of angular momentum.

– Particle motion with one degree of freedom.

– Vector differential equations. Projectile examples; examples in three dimensions.

– Plane polar coordinates. Acceleration in polars. Motion under a central force, the u-equation. Properties of conics; Kepler's laws of planetary motion. Stability of motion.

– Examples from cylindrical geometry.

– Systems with transfer of mass.